

by_normal_form_q9 over GF(9)

July 15, 2026

The equation

The equation of the quartic curve is :

$$X_0^4 + X_1^4 + 2X_2^4 + 4X_0^3X_1 + 8X_0^3X_2 + 6X_0X_1^3 + 6X_1^3X_2 + 3X_0X_2^3 + 4X_1X_2^3$$

$$(1, 1, 2, 4, 8, 6, 6, 3, 4, 0, 0, 0, 0, 0)$$

$$X_0^4 + X_1^4 + 2X_2^4 + 4X_0^3X_1 + 8X_0^3X_2 + 6X_0X_1^3 + 6X_1^3X_2 + 3X_0X_2^3 + 4X_1X_2^3$$

General information

Number of bitangents	28
Number of points	28
Fullness	is full
Number of Kovalevski points	63
Bitangent line type (a_0, a_1, a_2)	$(0, 28, 0)$
Number of singular points	0

All points on the curve

The quartic has 28 points:

The points on the quartic curve are:

$$\begin{array}{lll}
0 : P_4 = (1, 1, 0) & 7 : P_{25} = (6, 1, 1) & 14 : P_{48} = (2, 4, 1) \\
1 : P_6 = (3, 1, 0) & 8 : P_{27} = (8, 1, 1) & 15 : P_{62} = (7, 5, 1) \\
2 : P_9 = (6, 1, 0) & 9 : P_{36} = (8, 2, 1) & 16 : P_{65} = (1, 6, 1) \\
3 : P_{10} = (7, 1, 0) & 10 : P_{37} = (0, 3, 1) & 17 : P_{66} = (2, 6, 1) \\
4 : P_{15} = (4, 0, 1) & 11 : P_{39} = (2, 3, 1) & 18 : P_{70} = (6, 6, 1) \\
5 : P_{22} = (3, 1, 1) & 12 : P_{43} = (6, 3, 1) & 19 : P_{72} = (8, 6, 1) \\
6 : P_{23} = (4, 1, 1) & 13 : P_{44} = (7, 3, 1) & 20 : P_{77} = (4, 7, 1)
\end{array}$$

$$\begin{array}{lll}
21 : P_{78} = (5, 7, 1) & 24 : P_{84} = (2, 8, 1) & 27 : P_{90} = (8, 8, 1) \\
22 : P_{79} = (6, 7, 1) & 25 : P_{86} = (4, 8, 1) & \\
23 : P_{80} = (7, 7, 1) & 26 : P_{89} = (7, 8, 1) &
\end{array}$$

The points by rank are: (4, 6, 9, 10, 15, 22, 23, 25, 27, 36, 37, 39, 43, 44, 48, 62, 65, 66, 70, 72, 77, 78, 79, 80, 84, 86, 89, 90)

The Kovalevski points are:

$$\begin{array}{l}
0 : P_1 = (0, 1, 0) = a_3 \cap a_6 \cap c_{34} \cap c_{46} \\
1 : P_2 = (0, 0, 1) = a_1 \cap a_3 \cap b_1 \cap b_3 \\
2 : P_3 = (1, 1, 1) = b_5 \cap b_6 \cap c_{45} \cap c_{46} \\
3 : P_5 = (2, 1, 0) = a_2 \cap a_4 \cap c_{25} \cap c_{45} \\
4 : P_7 = (4, 1, 0) = c_{23} \cap c_{26} \cap c_{35} \cap c_{56} \\
5 : P_8 = (5, 1, 0) = a_5 \cap b_1 \cap c_{24} \cap c_{36} \\
6 : P_{11} = (8, 1, 0) = b_4 \cap b_5 \cap c_{14} \cap c_{15} \\
7 : P_{12} = (1, 0, 1) = a_1 \cap a_4 \cap c_{16} \cap c_{46} \\
8 : P_{13} = (2, 0, 1) = a_1 \cap a_5 \cap c_{14} \cap c_{45} \\
9 : P_{14} = (3, 0, 1) = a_1 \cap a_6 \cap c_{15} \cap c_{56} \\
10 : P_{16} = (5, 0, 1) = a_1 \cap b_5 \cap c_{26} \cap c_{34} \\
11 : P_{17} = (6, 0, 1) = a_1 \cap b_4 \cap c_{25} \cap c_{36} \\
12 : P_{18} = (7, 0, 1) = a_1 \cap b_6 \cap c_{24} \cap c_{35} \\
13 : P_{19} = (8, 0, 1) = a_1 \cap a_2 \cap c_{13} \cap c_{23} \\
14 : P_{20} = (0, 1, 1) = a_3 \cap a_4 \cap c_{23} \cap c_{24} \\
15 : P_{21} = (2, 1, 1) = c_{15} \cap c_{16} \cap c_{35} \cap c_{36} \\
16 : P_{24} = (5, 1, 1) = b_1 \cap b_4 \cap c_{13} \cap c_{34} \\
17 : P_{26} = (7, 1, 1) = a_2 \cap b_3 \cap c_{14} \cap c_{56} \\
18 : P_{28} = (0, 2, 1) = a_3 \cap b_2 \cap c_{16} \cap c_{45} \\
19 : P_{29} = (1, 2, 1) = b_2 \cap b_4 \cap c_{26} \cap c_{46} \\
20 : P_{30} = (2, 2, 1) = a_4 \cap b_2 \cap c_{13} \cap c_{56} \\
21 : P_{31} = (3, 2, 1) = a_6 \cap b_2 \cap c_{14} \cap c_{35} \\
22 : P_{32} = (4, 2, 1) = b_2 \cap b_6 \cap c_{23} \cap c_{36} \\
23 : P_{33} = (5, 2, 1) = b_2 \cap b_3 \cap c_{24} \cap c_{34} \\
24 : P_{34} = (6, 2, 1) = a_2 \cap a_5 \cap b_2 \cap b_5 \\
25 : P_{35} = (7, 2, 1) = b_1 \cap b_2 \cap c_{15} \cap c_{25} \\
26 : P_{38} = (1, 3, 1) = c_{13} \cap c_{14} \cap c_{36} \cap c_{46} \\
27 : P_{40} = (3, 3, 1) = a_5 \cap a_6 \cap c_{25} \cap c_{26} \\
28 : P_{41} = (4, 3, 1) = b_1 \cap b_5 \cap c_{16} \cap c_{56} \\
29 : P_{42} = (5, 3, 1) = a_2 \cap b_6 \cap c_{15} \cap c_{34} \\
30 : P_{45} = (8, 3, 1) = b_3 \cap b_4 \cap c_{35} \cap c_{45} \\
31 : P_{46} = (0, 4, 1) = a_3 \cap b_5 \cap c_{35} \cap d \\
32 : P_{47} = (1, 4, 1) = c_{15} \cap c_{23} \cap c_{46} \cap d \\
33 : P_{49} = (3, 4, 1) = a_6 \cap b_3 \cap c_{36} \cap d \\
34 : P_{50} = (4, 4, 1) = a_2 \cap b_4 \cap c_{24} \cap d \\
35 : P_{51} = (5, 4, 1) = c_{16} \cap c_{25} \cap c_{34} \cap d \\
36 : P_{52} = (6, 4, 1) = a_4 \cap b_1 \cap c_{14} \cap d \\
37 : P_{53} = (7, 4, 1) = c_{13} \cap c_{26} \cap c_{45} \cap d \\
38 : P_{54} = (8, 4, 1) = a_5 \cap b_6 \cap c_{56} \cap d
\end{array}$$

39 : $P_{55} = (0, 5, 1) = a_3 \cap b_4 \cap c_{12} \cap c_{56}$
 40 : $P_{56} = (1, 5, 1) = a_5 \cap b_3 \cap c_{12} \cap c_{46}$
 41 : $P_{57} = (2, 5, 1) = b_1 \cap b_6 \cap c_{12} \cap c_{26}$
 42 : $P_{58} = (3, 5, 1) = a_2 \cap a_6 \cap c_{12} \cap c_{16}$
 43 : $P_{59} = (4, 5, 1) = c_{12} \cap c_{13} \cap c_{25} \cap c_{35}$
 44 : $P_{60} = (5, 5, 1) = c_{12} \cap c_{14} \cap c_{23} \cap c_{34}$
 45 : $P_{61} = (6, 5, 1) = c_{12} \cap c_{15} \cap c_{24} \cap c_{45}$
 46 : $P_{63} = (8, 5, 1) = a_4 \cap b_5 \cap c_{12} \cap c_{36}$
 47 : $P_{64} = (0, 6, 1) = a_3 \cap b_6 \cap c_{14} \cap c_{25}$
 48 : $P_{67} = (3, 6, 1) = a_6 \cap b_5 \cap c_{13} \cap c_{24}$
 49 : $P_{68} = (4, 6, 1) = a_4 \cap b_3 \cap c_{15} \cap c_{26}$
 50 : $P_{69} = (5, 6, 1) = c_{34} \cap c_{36} \cap c_{45} \cap c_{56}$
 51 : $P_{71} = (7, 6, 1) = a_5 \cap b_4 \cap c_{16} \cap c_{23}$
 52 : $P_{73} = (0, 7, 1) = a_3 \cap a_5 \cap c_{13} \cap c_{15}$
 53 : $P_{74} = (1, 7, 1) = a_2 \cap b_1 \cap c_{35} \cap c_{46}$
 54 : $P_{75} = (2, 7, 1) = b_3 \cap b_5 \cap c_{23} \cap c_{25}$
 55 : $P_{76} = (3, 7, 1) = a_4 \cap a_6 \cap b_4 \cap b_6$
 56 : $P_{81} = (8, 7, 1) = c_{14} \cap c_{16} \cap c_{24} \cap c_{26}$
 57 : $P_{82} = (0, 8, 1) = a_2 \cap a_3 \cap c_{26} \cap c_{36}$
 58 : $P_{83} = (1, 8, 1) = c_{24} \cap c_{25} \cap c_{46} \cap c_{56}$
 59 : $P_{85} = (3, 8, 1) = a_6 \cap b_1 \cap c_{23} \cap c_{45}$
 60 : $P_{87} = (5, 8, 1) = a_4 \cap a_5 \cap c_{34} \cap c_{35}$
 61 : $P_{88} = (6, 8, 1) = b_3 \cap b_6 \cap c_{13} \cap c_{16}$
 62 : $P_0 = (1, 0, 0) = a_1 \cap b_2 \cap c_{12} \cap d$

The incidence structure induced by the Kovalevski points is:

Flags=(1, 7, 8, 9, 10, 11, 12, 13, 62, 66, 76, 80, 87, 92, 97, 105, 116, 120, 126,
 127, 140, 144, 157, 165, 173, 178, 183, 192, 196, 203, 209, 225, 235, 238, 244,
 249, 257, 260, 276, 279, 290, 292, 303, 304, 312, 315, 324, 336, 342, 348, 357,
 363, 370, 374, 379, 383, 394, 403, 406, 414, 419, 431, 437, 459, 460, 461, 462,
 463, 464, 465, 466, 503, 505, 521, 527, 534, 537, 544, 553, 558, 565, 573, 578,
 583, 586, 597, 601, 606, 618, 622, 632, 636, 640, 654, 658, 661, 676, 678, 684,
 695, 705, 715, 722, 731, 734, 740, 748, 754, 795, 796, 797, 798, 799, 800, 801,
 802, 818, 832, 835, 839, 845, 856, 862, 867, 871, 880, 888, 890, 899, 903, 908,
 918, 926, 929, 938, 951, 954, 960, 970, 974, 977, 990, 994, 997, 1015, 1023, 1026,
 1036, 1043, 1050, 1059, 1064, 1069, 1075, 1084, 1085, 1093, 1103, 1115, 1122,
 1125, 1130, 1139, 1146, 1148, 1157, 1168, 1179, 1182, 1190, 1192, 1200, 1208,
 1222, 1224, 1232, 1240, 1244, 1251, 1255, 1264, 1270, 1279, 1287, 1297, 1301,
 1309, 1316, 1317, 1323, 1333, 1339, 1346, 1352, 1358, 1367, 1373, 1383, 1390,
 1398, 1401, 1407, 1416, 1417, 1429, 1439, 1446, 1454, 1460, 1464, 1471, 1475,
 1482, 1495, 1499, 1506, 1514, 1515, 1520, 1530, 1542, 1549, 1557, 1562, 1571,
 1575, 1577, 1582, 1594, 1601, 1607, 1615, 1628, 1633, 1642, 1647, 1655, 1658,
 1666, 1676, 1677, 1688, 1696, 1732, 1733, 1734, 1735, 1736, 1737, 1738, 1739,
 1763)

nb_rows=28

nb_cols=63

nb_flags=252

[illegible]

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)
(0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,
0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0)
)
(1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
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)
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0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)
)
(0,
1, 1, 1, 1, 1, 1, 1, 0, 1)
)

The Kovalevski points sorted by rank are: (0, 1, 2, 3, 5, 7, 8, 11, 12, 13, 14, 16,
17, 18, 19, 20, 21, 24, 26, 28, 29, 30, 31, 32, 33, 34, 35, 38, 40, 41, 42, 45, 46,
47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 63, 64, 67, 68, 69, 71, 73,
74, 75, 76, 81, 82, 83, 85, 87, 88)

The points off the curve are:

0 : $P_0 = (1, 0, 0)$	22 : $P_{31} = (3, 2, 1)$	44 : $P_{59} = (4, 5, 1)$
1 : $P_1 = (0, 1, 0)$	23 : $P_{32} = (4, 2, 1)$	45 : $P_{60} = (5, 5, 1)$
2 : $P_2 = (0, 0, 1)$	24 : $P_{33} = (5, 2, 1)$	46 : $P_{61} = (6, 5, 1)$
3 : $P_3 = (1, 1, 1)$	25 : $P_{34} = (6, 2, 1)$	47 : $P_{63} = (8, 5, 1)$
4 : $P_5 = (2, 1, 0)$	26 : $P_{35} = (7, 2, 1)$	48 : $P_{64} = (0, 6, 1)$
5 : $P_7 = (4, 1, 0)$	27 : $P_{38} = (1, 3, 1)$	49 : $P_{67} = (3, 6, 1)$
6 : $P_8 = (5, 1, 0)$	28 : $P_{40} = (3, 3, 1)$	50 : $P_{68} = (4, 6, 1)$
7 : $P_{11} = (8, 1, 0)$	29 : $P_{41} = (4, 3, 1)$	51 : $P_{69} = (5, 6, 1)$
8 : $P_{12} = (1, 0, 1)$	30 : $P_{42} = (5, 3, 1)$	52 : $P_{71} = (7, 6, 1)$
9 : $P_{13} = (2, 0, 1)$	31 : $P_{45} = (8, 3, 1)$	53 : $P_{73} = (0, 7, 1)$
10 : $P_{14} = (3, 0, 1)$	32 : $P_{46} = (0, 4, 1)$	54 : $P_{74} = (1, 7, 1)$
11 : $P_{16} = (5, 0, 1)$	33 : $P_{47} = (1, 4, 1)$	55 : $P_{75} = (2, 7, 1)$
12 : $P_{17} = (6, 0, 1)$	34 : $P_{49} = (3, 4, 1)$	56 : $P_{76} = (3, 7, 1)$
13 : $P_{18} = (7, 0, 1)$	35 : $P_{50} = (4, 4, 1)$	57 : $P_{81} = (8, 7, 1)$
14 : $P_{19} = (8, 0, 1)$	36 : $P_{51} = (5, 4, 1)$	58 : $P_{82} = (0, 8, 1)$
15 : $P_{20} = (0, 1, 1)$	37 : $P_{52} = (6, 4, 1)$	59 : $P_{83} = (1, 8, 1)$
16 : $P_{21} = (2, 1, 1)$	38 : $P_{53} = (7, 4, 1)$	60 : $P_{85} = (3, 8, 1)$
17 : $P_{24} = (5, 1, 1)$	39 : $P_{54} = (8, 4, 1)$	61 : $P_{87} = (5, 8, 1)$
18 : $P_{26} = (7, 1, 1)$	40 : $P_{55} = (0, 5, 1)$	62 : $P_{88} = (6, 8, 1)$
19 : $P_{28} = (0, 2, 1)$	41 : $P_{56} = (1, 5, 1)$	
20 : $P_{29} = (1, 2, 1)$	42 : $P_{57} = (2, 5, 1)$	
21 : $P_{30} = (2, 2, 1)$	43 : $P_{58} = (3, 5, 1)$	

(0, 1, 2, 3, 5, 7, 8, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 24, 26, 28, 29, 30, 31, 32, 33, 34, 35, 38, 40, 41, 42, 45, 46, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 63, 64, 67, 68, 69, 71, 73, 74, 75, 76, 81, 82, 83, 85, 87, 88)

Duals of Bitangents

dual of bitangents:

- 0 : $P_1 = (0, 1, 0)$
- 1 : $P_{40} = (3, 3, 1)$
- 2 : $P_0 = (1, 0, 0)$
- 3 : $P_{30} = (2, 2, 1)$
- 4 : $P_{74} = (1, 7, 1)$
- 5 : $P_{19} = (8, 0, 1)$
- 6 : $P_8 = (5, 1, 0)$
- 7 : $P_{20} = (0, 1, 1)$
- 8 : $P_{10} = (7, 1, 0)$
- 9 : $P_{59} = (4, 5, 1)$
- 10 : $P_{69} = (5, 6, 1)$
- 11 : $P_{53} = (7, 4, 1)$
- 12 : $P_{55} = (0, 5, 1)$
- 13 : $P_{76} = (3, 7, 1)$
- 14 : $P_{47} = (1, 4, 1)$

- 15 : $P_{81} = (8, 7, 1)$
- 16 : $P_{21} = (2, 1, 1)$
- 17 : $P_{31} = (3, 2, 1)$
- 18 : $P_{35} = (7, 2, 1)$
- 19 : $P_{50} = (4, 4, 1)$
- 20 : $P_{42} = (5, 3, 1)$
- 21 : $P_{16} = (5, 0, 1)$
- 22 : $P_{71} = (7, 6, 1)$
- 23 : $P_{41} = (4, 3, 1)$
- 24 : $P_3 = (1, 1, 1)$
- 25 : $P_{13} = (2, 0, 1)$
- 26 : $P_{63} = (8, 5, 1)$
- 27 : $P_{64} = (0, 6, 1)$

Dimension of the vanishing ideal is 1

$$3*X2^4 + 3*X0^3*X1 + X0^3*X2 + 3*X0*X1^3 + 7*X1^3*X2 + 5*X0*X2^3 + 2*X1*X2^3$$

The lines and their points of contact are:

$$\begin{aligned}
a_1 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}_9 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}_9 P_{15} = \mathbf{P}(4, 0, 1) \ 4\times \\
a_2 &= \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 6 \end{bmatrix}_{66} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 6 \end{bmatrix}_{66} P_{66} = \mathbf{P}(2, 6, 1) \ 4\times \\
a_3 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{90} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{90} P_{37} = \mathbf{P}(0, 3, 1) \ 4\times \\
a_4 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{11} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{11} P_{44} = \mathbf{P}(7, 3, 1) \ 4\times \\
a_5 &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{25} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{25} P_{23} = \mathbf{P}(4, 1, 1) \ 4\times \\
a_6 &= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \end{bmatrix}_{40} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \end{bmatrix}_{40} P_{22} = \mathbf{P}(3, 1, 1) \ 4\times \\
b_1 &= \begin{bmatrix} 1 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{79} = \begin{bmatrix} 1 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{79} P_{72} = \mathbf{P}(8, 6, 1) \ 4\times \\
b_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}_2 P_{36} = \mathbf{P}(8, 2, 1) \ 4\times \\
b_3 &= \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{59} = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{59} P_{10} = \mathbf{P}(7, 1, 0) \ 4\times \\
b_4 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 7 \end{bmatrix}_{87} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 7 \end{bmatrix}_{87} P_{84} = \mathbf{P}(2, 8, 1) \ 4\times \\
b_5 &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{73} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{73} P_{89} = \mathbf{P}(7, 8, 1) \ 4\times \\
b_6 &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 8 \end{bmatrix}_{58} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 8 \end{bmatrix}_{58} P_6 = \mathbf{P}(3, 1, 0) \ 4\times \\
c_{12} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{bmatrix}_7 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{bmatrix}_7 P_{62} = \mathbf{P}(7, 5, 1) \ 4\times
\end{aligned}$$

$$\begin{aligned}
c_{13} &= \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 5 \end{bmatrix}_{65} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 5 \end{bmatrix}_{65} & P_9 = \mathbf{P}(6, 1, 0) \ 4\times \\
c_{14} &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 8 \end{bmatrix}_{28} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 8 \end{bmatrix}_{28} & P_{86} = \mathbf{P}(4, 8, 1) \ 4\times \\
c_{15} &= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 5 \end{bmatrix}_{45} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 5 \end{bmatrix}_{45} & P_{90} = \mathbf{P}(8, 8, 1) \ 4\times \\
c_{16} &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}_{12} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}_{12} & P_4 = \mathbf{P}(1, 1, 0) \ 4\times \\
c_{23} &= \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 1 \end{bmatrix}_{61} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 1 \end{bmatrix}_{61} & P_{43} = \mathbf{P}(6, 3, 1) \ 4\times \\
c_{24} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}_{51} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}_{51} & P_{39} = \mathbf{P}(2, 3, 1) \ 4\times \\
c_{25} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 8 \end{bmatrix}_{88} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 8 \end{bmatrix}_{88} & P_{27} = \mathbf{P}(8, 1, 1) \ 4\times \\
c_{26} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{76} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{76} & P_{25} = \mathbf{P}(6, 1, 1) \ 4\times \\
c_{34} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \end{bmatrix}_{70} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \end{bmatrix}_{70} & P_{78} = \mathbf{P}(5, 7, 1) \ 4\times \\
c_{35} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \end{bmatrix}_{53} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \end{bmatrix}_{53} & P_{70} = \mathbf{P}(6, 6, 1) \ 4\times \\
c_{36} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 6 \end{bmatrix}_{86} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 6 \end{bmatrix}_{86} & P_{80} = \mathbf{P}(7, 7, 1) \ 4\times \\
c_{45} &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}_{22} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}_{22} & P_{77} = \mathbf{P}(4, 7, 1) \ 4\times \\
c_{46} &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_{10} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_{10} & P_{65} = \mathbf{P}(1, 6, 1) \ 4\times \\
c_{56} &= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 7 \end{bmatrix}_{47} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 7 \end{bmatrix}_{47} & P_{79} = \mathbf{P}(6, 7, 1) \ 4\times \\
d &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \end{bmatrix}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \end{bmatrix}_3 & P_{48} = \mathbf{P}(2, 4, 1) \ 4\times
\end{aligned}$$

Rank of lines: (9, 66, 90, 11, 25, 40, 79, 2, 59, 87, 73, 58, 7, 65, 28, 45, 12, 61, 51, 88, 76, 70, 53, 86, 22, 10, 47, 3)

Line type: 1^{28}

28	1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 0
----	---	--

point types: 1^{28}

28	1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 0
----	---	--

point types for points off the curve: 4^{63}

63	4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 0
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Lines on points off the curve:

Off point $0 = P_0 = (1, 0, 0)$ lies on 4 bisecants : $\{ 0, 7, 12, 27 \}$
 Off point $1 = P_1 = (0, 1, 0)$ lies on 4 bisecants : $\{ 2, 5, 21, 25 \}$
 Off point $2 = P_2 = (0, 0, 1)$ lies on 4 bisecants : $\{ 0, 2, 6, 8 \}$
 Off point $3 = P_3 = (1, 1, 1)$ lies on 4 bisecants : $\{ 10, 11, 24, 25 \}$
 Off point $4 = P_5 = (2, 1, 0)$ lies on 4 bisecants : $\{ 1, 3, 19, 24 \}$
 Off point $5 = P_7 = (4, 1, 0)$ lies on 4 bisecants : $\{ 17, 20, 22, 26 \}$
 Off point $6 = P_8 = (5, 1, 0)$ lies on 4 bisecants : $\{ 4, 6, 18, 23 \}$
 Off point $7 = P_{11} = (8, 1, 0)$ lies on 4 bisecants : $\{ 9, 10, 14, 15 \}$
 Off point $8 = P_{12} = (1, 0, 1)$ lies on 4 bisecants : $\{ 0, 3, 16, 25 \}$
 Off point $9 = P_{13} = (2, 0, 1)$ lies on 4 bisecants : $\{ 0, 4, 14, 24 \}$
 Off point $10 = P_{14} = (3, 0, 1)$ lies on 4 bisecants : $\{ 0, 5, 15, 26 \}$
 Off point $11 = P_{16} = (5, 0, 1)$ lies on 4 bisecants : $\{ 0, 10, 20, 21 \}$
 Off point $12 = P_{17} = (6, 0, 1)$ lies on 4 bisecants : $\{ 0, 9, 19, 23 \}$
 Off point $13 = P_{18} = (7, 0, 1)$ lies on 4 bisecants : $\{ 0, 11, 18, 22 \}$
 Off point $14 = P_{19} = (8, 0, 1)$ lies on 4 bisecants : $\{ 0, 1, 13, 17 \}$
 Off point $15 = P_{20} = (0, 1, 1)$ lies on 4 bisecants : $\{ 2, 3, 17, 18 \}$
 Off point $16 = P_{21} = (2, 1, 1)$ lies on 4 bisecants : $\{ 15, 16, 22, 23 \}$
 Off point $17 = P_{24} = (5, 1, 1)$ lies on 4 bisecants : $\{ 6, 9, 13, 21 \}$
 Off point $18 = P_{26} = (7, 1, 1)$ lies on 4 bisecants : $\{ 1, 8, 14, 26 \}$
 Off point $19 = P_{28} = (0, 2, 1)$ lies on 4 bisecants : $\{ 2, 7, 16, 24 \}$
 Off point $20 = P_{29} = (1, 2, 1)$ lies on 4 bisecants : $\{ 7, 9, 20, 25 \}$
 Off point $21 = P_{30} = (2, 2, 1)$ lies on 4 bisecants : $\{ 3, 7, 13, 26 \}$
 Off point $22 = P_{31} = (3, 2, 1)$ lies on 4 bisecants : $\{ 5, 7, 14, 22 \}$
 Off point $23 = P_{32} = (4, 2, 1)$ lies on 4 bisecants : $\{ 7, 11, 17, 23 \}$
 Off point $24 = P_{33} = (5, 2, 1)$ lies on 4 bisecants : $\{ 7, 8, 18, 21 \}$
 Off point $25 = P_{34} = (6, 2, 1)$ lies on 4 bisecants : $\{ 1, 4, 7, 10 \}$
 Off point $26 = P_{35} = (7, 2, 1)$ lies on 4 bisecants : $\{ 6, 7, 15, 19 \}$
 Off point $27 = P_{38} = (1, 3, 1)$ lies on 4 bisecants : $\{ 13, 14, 23, 25 \}$
 Off point $28 = P_{40} = (3, 3, 1)$ lies on 4 bisecants : $\{ 4, 5, 19, 20 \}$
 Off point $29 = P_{41} = (4, 3, 1)$ lies on 4 bisecants : $\{ 6, 10, 16, 26 \}$
 Off point $30 = P_{42} = (5, 3, 1)$ lies on 4 bisecants : $\{ 1, 11, 15, 21 \}$
 Off point $31 = P_{45} = (8, 3, 1)$ lies on 4 bisecants : $\{ 8, 9, 22, 24 \}$
 Off point $32 = P_{46} = (0, 4, 1)$ lies on 4 bisecants : $\{ 2, 10, 22, 27 \}$
 Off point $33 = P_{47} = (1, 4, 1)$ lies on 4 bisecants : $\{ 15, 17, 25, 27 \}$
 Off point $34 = P_{49} = (3, 4, 1)$ lies on 4 bisecants : $\{ 5, 8, 23, 27 \}$

Off point 35 = $P_{50} = (4, 4, 1)$ lies on 4 bisecants : $\{ 1, 9, 18, 27 \}$
 Off point 36 = $P_{51} = (5, 4, 1)$ lies on 4 bisecants : $\{ 16, 19, 21, 27 \}$
 Off point 37 = $P_{52} = (6, 4, 1)$ lies on 4 bisecants : $\{ 3, 6, 14, 27 \}$
 Off point 38 = $P_{53} = (7, 4, 1)$ lies on 4 bisecants : $\{ 13, 20, 24, 27 \}$
 Off point 39 = $P_{54} = (8, 4, 1)$ lies on 4 bisecants : $\{ 4, 11, 26, 27 \}$
 Off point 40 = $P_{55} = (0, 5, 1)$ lies on 4 bisecants : $\{ 2, 9, 12, 26 \}$
 Off point 41 = $P_{56} = (1, 5, 1)$ lies on 4 bisecants : $\{ 4, 8, 12, 25 \}$
 Off point 42 = $P_{57} = (2, 5, 1)$ lies on 4 bisecants : $\{ 6, 11, 12, 20 \}$
 Off point 43 = $P_{58} = (3, 5, 1)$ lies on 4 bisecants : $\{ 1, 5, 12, 16 \}$
 Off point 44 = $P_{59} = (4, 5, 1)$ lies on 4 bisecants : $\{ 12, 13, 19, 22 \}$
 Off point 45 = $P_{60} = (5, 5, 1)$ lies on 4 bisecants : $\{ 12, 14, 17, 21 \}$
 Off point 46 = $P_{61} = (6, 5, 1)$ lies on 4 bisecants : $\{ 12, 15, 18, 24 \}$
 Off point 47 = $P_{63} = (8, 5, 1)$ lies on 4 bisecants : $\{ 3, 10, 12, 23 \}$
 Off point 48 = $P_{64} = (0, 6, 1)$ lies on 4 bisecants : $\{ 2, 11, 14, 19 \}$
 Off point 49 = $P_{67} = (3, 6, 1)$ lies on 4 bisecants : $\{ 5, 10, 13, 18 \}$
 Off point 50 = $P_{68} = (4, 6, 1)$ lies on 4 bisecants : $\{ 3, 8, 15, 20 \}$
 Off point 51 = $P_{69} = (5, 6, 1)$ lies on 4 bisecants : $\{ 21, 23, 24, 26 \}$
 Off point 52 = $P_{71} = (7, 6, 1)$ lies on 4 bisecants : $\{ 4, 9, 16, 17 \}$
 Off point 53 = $P_{73} = (0, 7, 1)$ lies on 4 bisecants : $\{ 2, 4, 13, 15 \}$
 Off point 54 = $P_{74} = (1, 7, 1)$ lies on 4 bisecants : $\{ 1, 6, 22, 25 \}$
 Off point 55 = $P_{75} = (2, 7, 1)$ lies on 4 bisecants : $\{ 8, 10, 17, 19 \}$
 Off point 56 = $P_{76} = (3, 7, 1)$ lies on 4 bisecants : $\{ 3, 5, 9, 11 \}$
 Off point 57 = $P_{81} = (8, 7, 1)$ lies on 4 bisecants : $\{ 14, 16, 18, 20 \}$
 Off point 58 = $P_{82} = (0, 8, 1)$ lies on 4 bisecants : $\{ 1, 2, 20, 23 \}$
 Off point 59 = $P_{83} = (1, 8, 1)$ lies on 4 bisecants : $\{ 18, 19, 25, 26 \}$
 Off point 60 = $P_{85} = (3, 8, 1)$ lies on 4 bisecants : $\{ 5, 6, 17, 24 \}$
 Off point 61 = $P_{87} = (5, 8, 1)$ lies on 4 bisecants : $\{ 3, 4, 21, 22 \}$
 Off point 62 = $P_{88} = (6, 8, 1)$ lies on 4 bisecants : $\{ 8, 11, 13, 16 \}$

The gradient

The gradient of the quartic curve is :

$$\begin{aligned}
 &X_0^3 + 6X_1^3 + 3X_2^3 \\
 &(1, 6, 3, 0, 0, 0, 0, 0, 0) \\
 &4X_0^3 + X_1^3 + 4X_2^3 \\
 &(4, 1, 4, 0, 0, 0, 0, 0, 0) \\
 &8X_0^3 + 6X_1^3 + 2X_2^3 \\
 &(8, 6, 2, 0, 0, 0, 0, 0, 0)
 \end{aligned}$$